

MYTHS AND STORIES

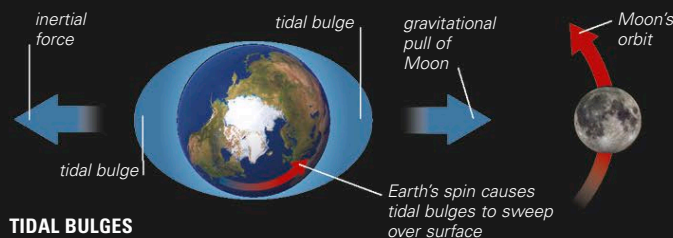
WEREWOLVES

Many myths and old folk tales attribute strange powers to the Moon. Some say that a full Moon can turn people mad (the origin of the word “lunacy”), and many cultures, from Eurasia to the Americas, share a belief that when the Moon is full, some humans can be transformed into vicious werewolves. The superstition is widespread and ancient—even the Babylonian King Nebuchadnezzar (c. 630–c. 562 BCE) imagined that he had become a werewolf.



LUNAR INFLUENCES

Although the Moon is much smaller than Earth, its gravity still exerts an influence. The Moon's gravitational attraction is felt most strongly on the side of Earth facing the Moon, and this pulls water in the oceans toward it. Inertia (the tendency of objects with mass to resist forces acting upon them) attempts to keep the water in place, but because the gravitational force is greater, a bulge of water is pulled toward the Moon. On the opposite side of Earth, the water's inertia is stronger than the Moon's gravity, so a second bulge of water is created. As Earth rotates, the bulges sweep over the planet's surface, creating daily changes in sea level called tides. The time of the high tide changes according to the Moon's position in the sky. The height of the tides changes during the lunar cycle, but the actual height also depends on local geography. In shallow coastal bays, the tidal range can be huge.



TIDAL BULGES

Gravitational interaction between Earth and the Moon creates two bulges in Earth's oceans (exaggerated here). As Earth spins on its axis, the bulges of water sweep over the surface, creating tides.

TIDAL RANGE

The magenta in this satellite image of Morecambe Bay on the northwest coast of England reveals the inlets and mud flats that are left exposed at low tide.

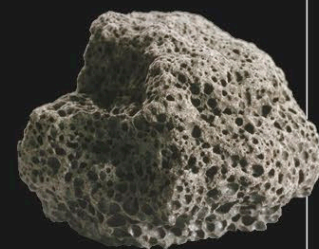


LAVA TUBE

Over 3 miles (5 km) wide and hundreds of miles long, this rille is a collapsed tubelike structure through which lava once flowed. Moonquakes caused by nearby impacts may have caused the roof to fall in.

SURFACE FEATURES

The surface of the Moon has been pulverized by meteorites and is covered by a rough, porous blanket of rubble several yards thick. This debris ranges in size from particles of dust to huge lumps of rock dozens of yards across. The soil (or regolith) consists of fine-grained, fragmented bedrock, the size of the grains getting progressively larger with depth. Because there is no wind or rain, the surface material does not move far, and its composition can change considerably from place to place. The thickness also varies—in young mare regions, it is about 16 ft (5 m) thick, but this increases to 32 ft (10 m) in the old highlands. Micrometeorite impacts continuously erode exposed rocks, and they are also damaged by cosmic rays and solar-flare particles. The topmost layer of soil is saturated with hydrogen ions absorbed from the solar wind.



MOON ROCK

This 6-in (15-cm) wide rock formed as lava from the interior rose to the Moon's surface and solidified. The small holes were formed as gas bubbles escaped.

TRACKS IN THE SOIL

Lunar Rover tire tracks lead away from the Apollo 15 module “Falcon,” nestling near Hadley Rille in 1971. Over a million or more years, they will eventually be erased by meteorite bombardment.